

Computing Quasi-Normal Modes of
Schwarzschild Black Holes
with Physics-Informed Neural Networks
Supervisor Progress Update

Jonathan Chung

University of Cambridge

18 March 2026

Outline

- 1 Recap
- 2 Greedy Adaptive Resampling
- 3 Curriculum Learning
- 4 Inverse Problem: Learning Physics from Data
- 5 Comprehensive Results
- 6 Summary & Outlook

The Problem — Black Hole Quasi-Normal Modes

A perturbed Schwarzschild black hole rings down via damped sinusoids (QNMs):

$$\Phi(t) \propto e^{-t/\tau} \cos(\omega t)$$

Linearising the Einstein equations yields a 1+1D wave equation:

$$-\frac{\partial^2 \Phi}{\partial t^2} + \frac{\partial^2 \Phi}{\partial x^2} - V_\ell(r) \Phi = 0$$

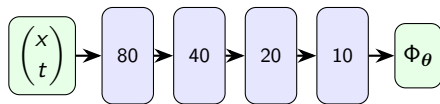
- $x = r + 2M \ln(\frac{r}{2M} - 1)$ is the tortoise coordinate.
- We focus on the **Zerilli potential**, $\ell = 2$ (dominant mode).
- Domain: $x/M \in [-50, 150]$, $t/M \in [0, 50]$.
- **Goal:** Solve numerically and extract $\omega M = 0.3737$, $\tau/M = 11.241$ (Leaver 1985).

Our Approach — Physics-Informed Neural Networks

Train a neural network $\Phi_{\theta}(x, t)$ to satisfy PDE + BCs + ICs:

$$\mathcal{L} = \sum_{i=1}^7 \lambda_i \mathcal{L}_i \quad (\text{PDE, BCs, ICs})$$

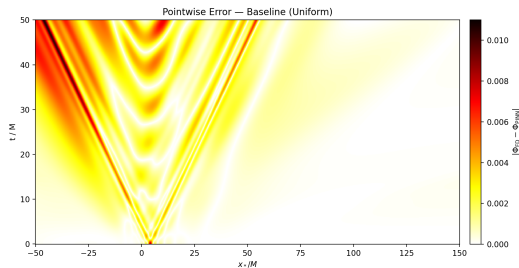
- **Architecture:** [2, 80, 40, 20, 10, 1], tanh, 4 521 params.
- **Training:** Adam 10k $\rightarrow$ L-BFGS 15k.
- **Points:** 33 200 collocation (vs. $\sim 501\,000$ FD grid).
- **QNM extraction:**
M1 (FFT + envelope) and M2 (curve fit) at $x_q = 10M$.



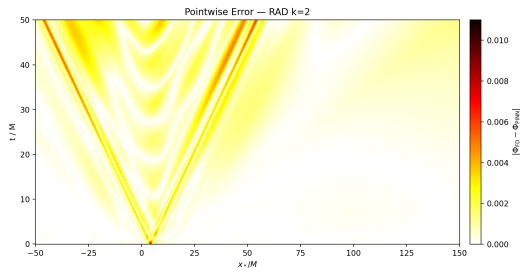
Output transform: $A \tanh(\text{raw})$

Where We Were (Last Meeting — 25 Feb)

Baseline (Uniform Sampling)



RAD $k=2$ (Best at Last Meeting)



	RMSD	M1 τ err	M2 τ err
Uniform baseline	0.00183	5.07%	1.50%
RAD $k=2$	0.00095 (-48%)	3.65%	1.80%

RAD improved **spatial accuracy** but QNM extraction remained similar.

Today: Three new strategies to push further.

Greedy Resampling — Algorithm

Instead of **probabilistic** sampling (RAD: $p_i \propto |r_i|^k + c$), use **deterministic top- N** selection:

- 1 Every P steps during Adam, generate $N_{\text{cand}} = 100,000$ random candidates.
- 2 Evaluate PDE residual $|r_i|$ at each candidate.
- 3 **Sort by $|r_i|$ descending** — keep the top $f \times N_r$ points.
- 4 Fill the remaining $(1-f) \times N_r$ slots with **uniform random** points.
- 5 Replace all $N_r = 32,000$ domain collocation points.

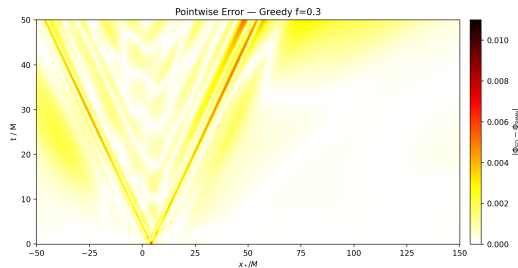
Key differences from RAD:

- **Deterministic** — no sampling noise; the most extreme residuals are always selected.
- **Single hyperparameter f** (fraction) — no tuning of k or c .
- $f = 0$: uniform; $f = 1$: fully greedy.

Greedy Resampling — Sweep Results

Configuration	RMSD	vs Baseline
Uniform (baseline)	0.00183	—
RAD $k=2$	0.00095	−48%
Greedy $f=0.3$	0.00085	−54%
Greedy $f=0.5$	0.00107	−42%
Greedy $f=0.8+P500$	0.00104	−43%

- $f=0.3$ is optimal: 30% greedy, 70% uniform.
- Too aggressive ($f \geq 0.5$): loses exploration.
- **Surpasses best RAD** with simpler algorithm.



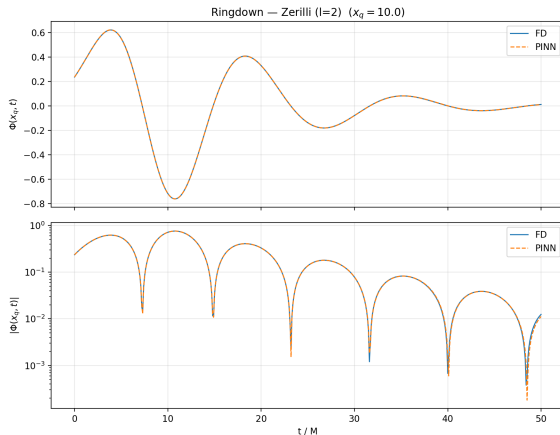
Error heatmap: Greedy $f=0.3$

Extended L-BFGS — Best Forward Model

Idea: Double L-BFGS iterations from 15k $\rightarrow$ 30k.

	RMSD	vs Base
Greedy $f=0.3$	0.00085	−54%
+ L-BFGS 30k	0.00082	−55%

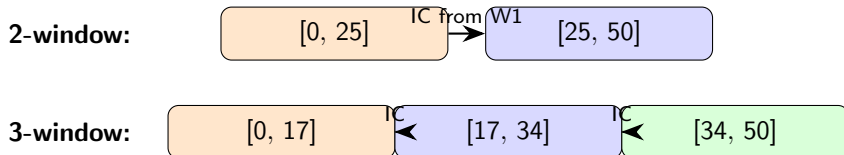
- **Best single-window forward model.**
- M1: ω err 0.68%, τ err 2.83%.
- M2: ω err 1.54%, τ err 1.99%.
- Marginal gain vs. $f=0.3$ alone — near convergence of the forward approach.



Ringdown at $x_q = 10M$: Greedy + L-BFGS 30k

Curriculum Learning — Strategy

Idea: Train on sequential time windows instead of the full domain.



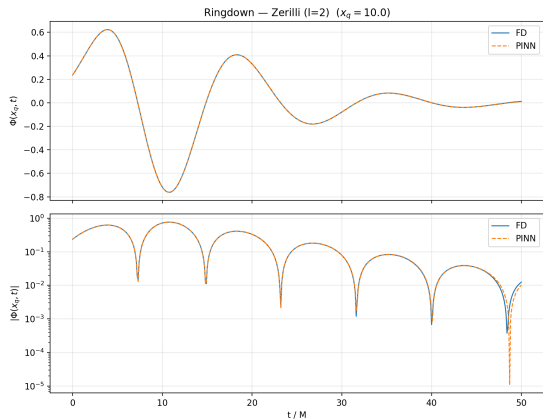
- Each window: independent PINN, full Adam + L-BFGS training.
- Previous window's trained output provides **initial condition** for the next.
- Uses **greedy** $f=0.3$ resampling within each window.
- Final solution: stitch all windows together.
- **Motivation:** Smaller time domains $\Rightarrow$ easier optimisation landscape.

Curriculum Learning — Results

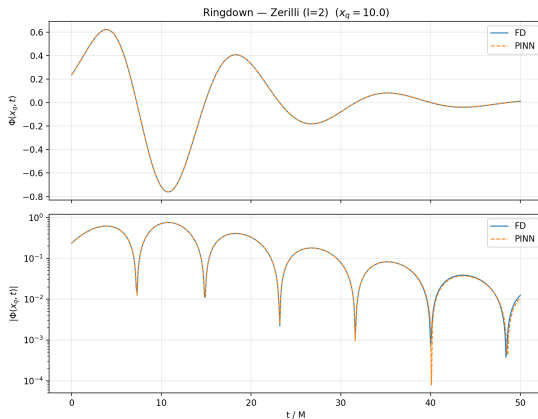
Model	RMSD	vs Base	M1 ω err	M1 τ err	M2 ω err	M2 τ err
Uniform baseline	0.00183	—	1.08%	5.07%	1.25%	1.50%
Greedy $f=0.3$	0.00085	−54%	0.68%	2.86%	1.54%	1.99%
Curriculum 2w	0.00085	−54%	0.70%	2.95%	1.53%	2.11%
Curriculum 3w	0.00098	−47%	0.69%	3.83%	1.59%	1.93%

- **2-window:** Competitive with single-window greedy (RMSD ≈ 0.00085).
- **3-window:** Slightly worse — error accumulates at window boundaries.
- Curriculum works but does **not outperform** the simpler greedy approach.
- IC transfer introduces interpolation error at each stitching boundary.

Curriculum Learning — Ringdown Comparison



2-window: $[0, 25] + [25, 50]$



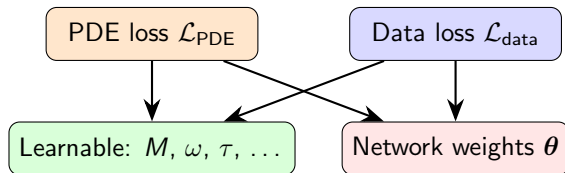
3-window: $[0, 17] + [17, 34] + [34, 50]$

Both match the FD waveform well at $x_q = 10M$.

A Different Approach — Inverse PINNs

Forward PINN: Given PDE parameters $\rightarrow$ solve for $\Phi \rightarrow$ extract QNMs.

Inverse PINN: Treat PDE parameters as **learnable** — optimise them alongside the network weights using PDE *and data* losses.



Total loss (Inverse PINN — learn M):

$$\mathcal{L} = \underbrace{\lambda_r \mathcal{L}_r + \lambda_{r_x} \mathcal{L}_{r_x} + \lambda_{r_t} \mathcal{L}_{r_t}}_{\text{PDE residual (same as forward)}} + \underbrace{\lambda_{ic} \mathcal{L}_{ic} + \lambda_{iv} \mathcal{L}_{iv}}_{\text{ICs}} + \underbrace{\lambda_{bl} \mathcal{L}_{bl} + \lambda_{br} \mathcal{L}_{br}}_{\text{BCs}} + \underbrace{\lambda_{\text{data}} \mathcal{L}_{\text{data}}}_{\text{new}}$$

- $\mathcal{L}_{\text{data}} = \text{MSE}(\Phi_{\text{PINN}}, \Phi_{\text{FD}})$ at 500 random (x, t) points. $\lambda_{\text{data}} = 10$.
- PDE loss now depends on M through $V(r)$ and $x(r) \Rightarrow$ gradients flow to M .

Inverse PINN — Learning Black Hole Mass M

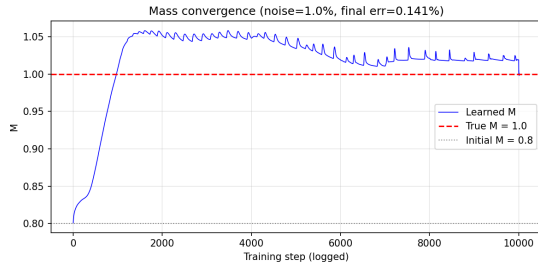
M enters through the Zerilli potential $V(r)$ and the tortoise coordinate $x(r)$.

Setup:

- Initialise $M = 0.80$ (true: $M = 1.0$).
- M is a trainable variable — optimised jointly with θ .
- Same training pipeline + data loss.

Result:

	Value
True M	1.0000
Learned M	0.9986
Error	0.14%



M converges from 0.80 to 0.9986

Inverse PINN + Ringdown Template — Concept

Learn simultaneously: M (via PDE) and ω, τ, A, ϕ_0 (via ringdown template).

Total loss:

$$\mathcal{L} = \underbrace{\lambda_r \mathcal{L}_r + \lambda_{r_x} \mathcal{L}_{r_x} + \lambda_{r_t} \mathcal{L}_{r_t}}_{\text{PDE residual (depends on } M)} + \underbrace{\lambda_{ic} \mathcal{L}_{ic} + \lambda_{iv} \mathcal{L}_{iv}}_{\text{ICs}} + \underbrace{\lambda_{bl} \mathcal{L}_{bl} + \lambda_{br} \mathcal{L}_{br}}_{\text{BCs}} + \underbrace{\lambda_{\text{data}} \mathcal{L}_{\text{data}}}_{\text{match fixed FD data}} + \underbrace{\lambda_{\text{ring}} \mathcal{L}_{\text{ring}}}_{\text{match learnable ringdown ansatz}}$$

- $\mathcal{L}_{\text{data}} = \text{MSE}(\Phi_{\text{PINN}}, \Phi_{\text{FD}}^{\text{noisy}})$ — compares against **fixed** noisy FD observations at 500 random (x, t) points across the full domain. $\lambda_{\text{data}} = 10$.
- $\mathcal{L}_{\text{ring}} = \text{MSE}(\Phi_{\text{PINN}}(x_0, t), A e^{-t/\tau} \cos(\omega t + \phi_0))$ — compares against a **learnable** damped-sinusoid template at one point $x_0=10$, late times $t \in [10, 50]$ only (200 pts). $\lambda_{\text{ring}} = 1$.
- M learned through $V(x; M)$ in PDE; ω, τ, A, ϕ_0 learned through $\mathcal{L}_{\text{ring}}$.

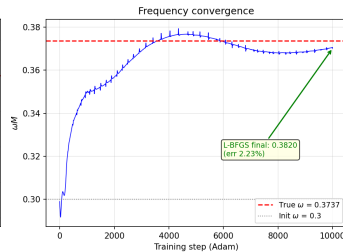
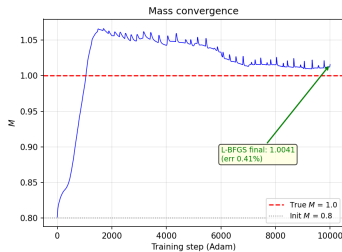
Inverse PINN + Ringdown Template — Results

Parameter recovery:

	Init	Learned	Err
M	0.80	1.004	0.41%
ωM	0.30	0.382	2.23%
τ/M	8.0	11.89	5.77%

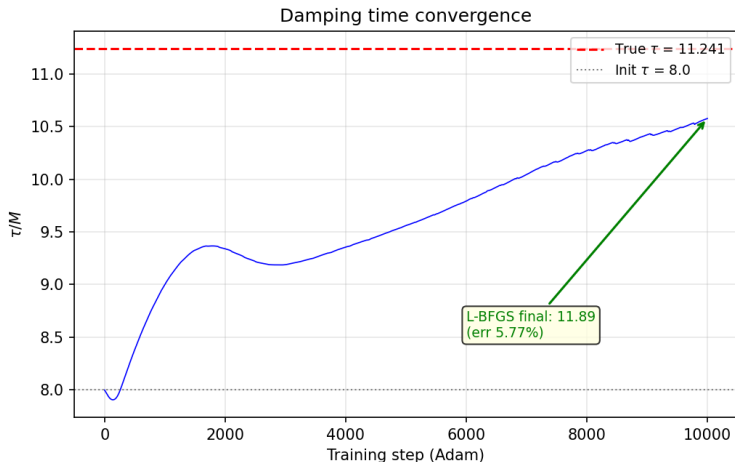
Key result:

Prony extraction from the waveform gives τ error = **1.52%**
— **best of all experiments!**



M and ω convergence from initial guesses. Green line marks Adam $\rightarrow$ L-BFGS transition.

Inverse PINN + Ringdown Template — Damping Time



τ convergence from initial guess $\tau_0 = 8.0$ to learned $\tau = 11.89$ (true: 11.24, error 5.77%). Green line marks Adam $\rightarrow$ L-BFGS transition.

Why Does the Template Loss Help Prony?

The problem with forward models:

- At late times, signal decays to $\sim 10^{-3}$ – 10^{-4} .
- PDE residual noise is $O(10^{-3})$ — comparable to the signal itself.
- Prony fits $\ln |\text{peaks}|$ vs. $t \Rightarrow$ noise corrupts the slope.
- **Result:** τ systematically underestimated by ~ 3 – 5% .

What the template loss does:

- Forces $\Phi \approx A e^{-t/\tau} \cos(\omega t + \phi_0)$ at late times.
- **Regularises** the exponential decay envelope.
- Prony reads a clean decay $\Rightarrow$ accurate τ .

Template Loss — Trade-off Between Extraction Methods

But why is Inverse PINN + Ringdown Template the worst on Method 2?

- The template loss overwrites PDE-derived physics with the **learned** (imperfect) parameters ω , τ , A , ϕ_0 .
- Method 2 (curve fit) reads those parameters *back* from the waveform — it recovers the learned values, not the true values.
- M2 τ error = 5.27% (worst of all experiments).

Prony $\leftrightarrow$ waveform cleanliness (late-time envelope shape)

Method 2 $\leftrightarrow$ PDE-encoded physics (overall waveform fidelity)

No single method dominates both extraction approaches.

RMSD Rankings — All Forward Models

Rank	Experiment	RMSD	vs Baseline
1	Greedy $f=0.3$ + L-BFGS 30k	0.000817	−55%
2	Curriculum 2-window	0.000845	−54%
3	Greedy $f=0.3$	0.000850	−54%
4	RAD $k=2$	0.000948	−48%
5	Curriculum 3-window	0.000976	−47%
6	Greedy $f=0.8$ + $P500$	0.001041	−43%
7	RAD $k=2$ + $P500$	0.001052	−43%
8	Greedy $f=0.5$	0.001066	−42%
9	RAD default	0.001150	−37%
10	RAD anchor	0.001412	−23%
11	Greedy $f=0.8$	0.001647	−10%
12	Paper baseline	0.001831	—

- All adaptive methods improve over uniform sampling.
- **Greedy $f=0.3$ + L-BFGS 30k**: best spatial accuracy.
- Inverse models (RMSD ~ 0.004 – 0.005) not shown — different objective.

Method 1 (Prony) Rankings — ω and τ Accuracy

	Experiment	ω err %	τ err %	Avg
1	Inverse PINN + Ringdown Template	1.02	1.52	1.27
2	RAD anchor	0.61	2.82	1.72
3	Greedy $f=0.3$ + L-BFGS 30k	0.68	2.83	1.76
4	Greedy $f=0.3$	0.68	2.86	1.77
5	Greedy $f=0.8$ + $P500$	0.69	2.85	1.77
6	RAD $k=2$ + $P500$	0.68	2.88	1.78
7	Curriculum 2-window	0.70	2.95	1.82
8	Greedy $f=0.5$	0.74	3.65	2.20
9	RAD $k=2$	0.85	3.65	2.25
10	Curriculum 3-window	0.69	3.83	2.26
11	RAD default	0.80	3.77	2.29
12	Greedy $f=0.8$	0.89	4.08	2.49
13	Paper baseline	1.08	5.07	3.08

- **Inverse PINN + Ringdown Template**: best Prony τ (1.52%) via template regularisation.
- Forward models cluster at 1.7–2.3% average error.

Method 2 (Curve Fit) Rankings — ω and τ Accuracy

	Experiment	ω err %	τ err %	Avg
1	Paper baseline	1.25	1.50	1.38
2	RAD default	1.44	1.65	1.55
3	Greedy $f=0.8$	1.42	1.80	1.61
4	RAD $k=2$	1.43	1.80	1.61
5	Greedy $f=0.5$	1.50	1.89	1.70
6	Curriculum 3-window	1.59	1.93	1.76
7	Greedy $f=0.3$	1.54	1.99	1.76
8	RAD anchor	1.57	1.96	1.77
9	Greedy $f=0.3$ + L-BFGS 30k	1.54	1.99	1.77
10	Curriculum 2-window	1.53	2.11	1.82
11	Greedy $f=0.8$ + $P500$	1.58	2.21	1.90
12	RAD $k=2$ + $P500$	1.57	2.24	1.90
13	Inverse PINN	1.66	2.49	2.07
14	Inverse PINN + Ringdown Template	2.18	5.27	3.72

- **Paper baseline leads M2** — template regularisation *hurts* curve fitting.
- Inverse PINN + Ringdown Template: best on M1 but **worst on M2** (template encodes imperfect learned params).

Key Takeaways

- 1 **Greedy $f=0.3$ + Extended L-BFGS:** Best spatial accuracy (RMSE -55%). Simple, effective, no complex hyperparameter tuning.
- 2 **Curriculum learning:** Competitive with greedy (2-window), but does not outperform it. Error accumulates with more windows.
- 3 **Inverse PINN with template loss:** Best Prony τ extraction (1.52% error). Template regularisation cleans the late-time waveform.
- 4 **No single method dominates all metrics.**
 - Best RMSE $\neq$ best QNM extraction.
 - Best M1 $\neq$ best M2.
 - The extraction method matters as much as the solver.

Future directions: Extend to Kerr black holes? Multi-mode ($\ell > 2$) extraction?

Thank you

Questions?

Appendix: QNM Extraction — Method 1 (FFT + Envelope)

Extract ω and τ **independently** from the late-time waveform $\Phi(x_q, t)$.

Step 1 — Estimate ω via FFT:

- 1 Window the signal; subtract the mean; apply Hann window.
- 2 Zero-pad by $64\times$ and compute the real FFT.
- 3 Peak frequency refined with **parabolic interpolation**.
- 4 $\omega = 2\pi f_{\text{peak}}$.

Step 2 — Estimate τ via log-linear envelope fit:

- 1 Find local maxima of $|\Phi(x_q, t)|$ (envelope peaks).
- 2 $\ln |y_{\text{peaks}}| = \ln A - t/\tau$.
- 3 Linear fit $\Rightarrow$ slope $m = -1/\tau$, hence $\tau = -1/m$.

Appendix: QNM Extraction — Method 2 (Nonlinear Curve Fit)

Fit the full damped-cosine model **simultaneously**:

$$\Phi(x_q, t) \approx A e^{-t/\tau} \cos(\omega t + \phi).$$

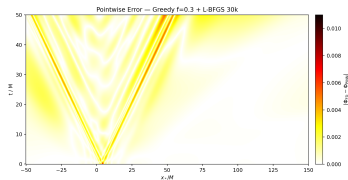
Procedure:

- 1 Initial guesses from Method 1: ω_0 , τ_0 , $A_0 = \max |\Phi|$, $\phi_0 = 0$.
- 2 `scipy.optimize.curve_fit` (Levenberg–Marquardt) with four free parameters.

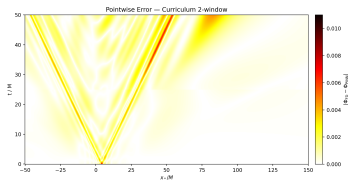
Advantages: All parameters fitted jointly; uses every data point.

Disadvantage: Requires good initial guesses; non-robust to noise.

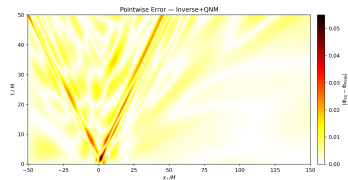
Appendix: Error Heatmap Gallery



Greedy $f=0.3$ + L-BFGS 30k



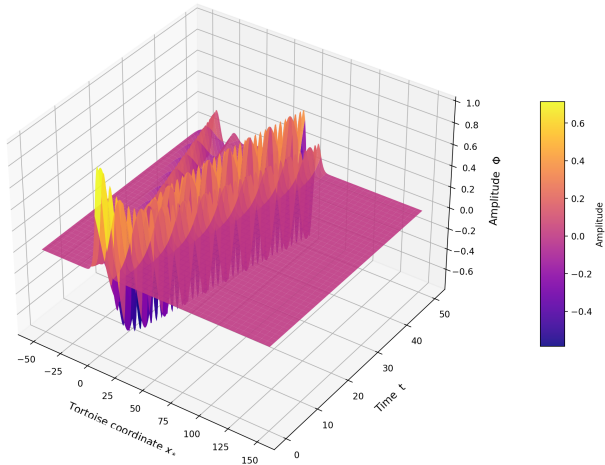
Curriculum 2-window



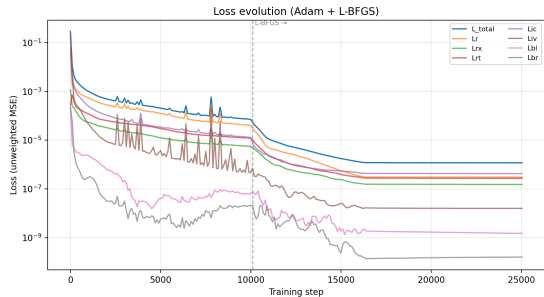
Inverse PINN + Ringdown Template

Appendix: 3D Waveform Visualisation

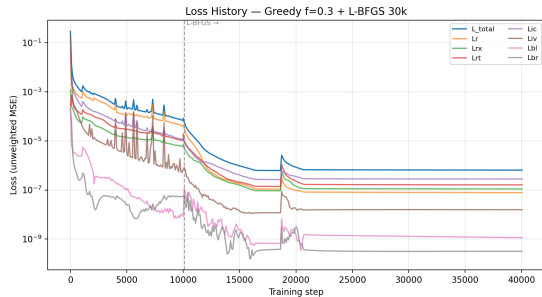
3D Waveform of Schwarzschild Black Hole Ringdown (FD Reference)



Appendix: Loss History — Baseline vs Greedy



Uniform baseline



Greedy $f=0.3$ + L-BFGS 30k